Spike / Feasibility — Offline Bangla Dubbing Video Player

Oct 29, 2026 · @Md. Nazmul Hossain

Verdict

Feasible on paper; needs one real run to confirm. The licence check found a fully commercial-safe offline stack. It also found a Bangladeshi Bangla speech corpus that reduces the biggest risk from the gap analysis.

- **Licences:** cleared. Whisper, pyannote 3.1, IndicTrans2 and Indic Parler-TTS all permit commercial use. NLLB-200 and MMS-TTS are excluded (BRD v1.1, NFR-08).
- **Bangladeshi voice:** SUBAK.KO, a 241-hour Bangladeshi Bangla corpus under CC-BY-4.0, makes an accent fine-tune possible. The BRD now treats accent (BR-04b) as phase 2, so this spike only scores the preset voices.
- **Pass conditions follow BRD v1.1:** 4 guaranteed speakers, 95% of lines within 0.5 s, English and Hindi guaranteed. Only E1 now blocks the project outright.
- **Still unproven:** translation naturalness, voice quality, diarization accuracy, speed and sync. These need the prototype run on real hardware.

The prototype code is written and syntax-checked, but it has not been run against the models yet, because they need a GPU machine with internet for the first download.

Scope, timebox and test setup

Timebox: 5 working days, one developer. Stop when the go / no-go rules below can be applied.

In the spike: the pre-process pipeline end to end, a minimal PyQt player, measurements.
Not in the spike: live mode, final UI, packaging, voice fine-tuning (only its feasibility).

Test machines

Machine	Purpose
PC with NVIDIA GPU, 8 GB+ VRAM	Recommended-hardware timings
Mac with Apple Silicon, 16 GB	macOS timings

Machine	Purpose
CPU-only laptop	Minimum-hardware timings

Test clips (5 minutes each)

Clip	Why
English, 1 speaker (lecture)	Baseline quality
English, 4 speakers (film or drama)	BR-05 guaranteed speaker count
Hindi, 2 speakers	Guaranteed source language (BR-02)
Korean, 2 speakers	Best-effort language via English pivot
English with loud music	Background separation and its fallback (BR-06)

Reviewers: at least 3 native Bangladeshi Bangla speakers score each output from 1 to 5.

Licence verification

Checked against model cards and repositories on 2026-10-29.

Component	Model	Licence found	Commercial	Notes
Speech-to-text	faster-whisper	MIT	Yes	Runs Whisper on CTranslate2
Diarization	pyannote speaker-diarization-3.1	MIT	Yes	Gated: accept conditions and use a Hugging Face token once
Translation	IndicTrans2	MIT	Yes	English ↔ Bengali (ben_Beng); Indic ↔ Indic
Translation	NLLB-200	CC-BY-NC 4.0	No	Research model; spike comparison only

Component	Model	Licence found	Commercial	Notes
Bangla TTS	Indic Parler-TTS	Apache 2.0	Yes	Gated download; 6 named Bengali voices (Arjun, Aditi, Tapan, Rashmi, Arnav, Riya)
Bangla TTS	bn-parler-tts	Not confirmed	Check	Community fine-tune on SUBAK.KO; no named speakers
Voice data	SUBAK.KO	CC-BY-4.0	Yes, with attribution	241 h Bangladeshi Bangla from SUST; 61 native speakers

Key findings

- IndicTrans2 is MIT, but it only translates between English and Indian languages. Korean, Arabic and other sources need a pivot: Whisper's built-in translate-to-English, then IndicTrans2 English → Bangla.
- Indic Parler-TTS's Bengali voices are trained mostly on Indian data. The SUBAK.KO speakers come from 8 divisions and 34 districts of Bangladesh, making it the best public base for a Bangladeshi voice.
- SUBAK.KO is an ASR corpus, so it needs cleaning before TTS fine-tuning; the bn-parler-tts project shows this route works.

Experiments

Fill the Result column during the spike run.

#	Experiment	Pass condition (BRD v1.1)	If it fails	Result
E1	End-to-end offline run	5-min English clip dubs into Bangla with the network off (after model download)	**Rework** to step 1	

#	Experiment	Pass condition (BRD v1.1)	If it fails	Result
E2	Translation naturalness	IndicTrans2 average reviewer score 3.5/5 or higher; local LLM scored for comparison	Use local LLM; rework only if both fail	
E3	Hindi and pivot languages	Hindi clip scores 3.5/5 or higher; Korean via English pivot scores 3/5 or higher	Hindi fail: tune or rework; Korean fail: stays best-effort	
E4	Preset voices	Score recorded for each of the 6 Indic Parler Bengali voices	Not blocking; informs BR-04b phase 2	
E5	Diarization	90%+ correct speaker labels on the 4-speaker clip	Tune once; BR-17 covers residual errors; rework if under 80%	
E6	Speed	Pre-process under 15 min per hour of video on recommended hardware (NFR-10); timings for all machines	Live mode moves to version 2	
E7	Sync	95% of lines within 0.5 s; no line over 1.0 s late (NFR-05)	Tune cap; BR-15 editor shortens lines	
E8	Background kept	Reviewers confirm music and effects remain audible	Use audio-lowering fallback (`--no-separation`)	
E9	Bangladeshi vocabulary	Reviewers rate wording 3.5/5 or higher; list words to add to the glossary (BR-04a)	Build glossary before development	

Prototype code

Delivered as **bangla-dub-spike.zip** in the chat. Two scripts cover experiments E1–E8.

File	What it does
`dub_pipeline.py`	Pre-process pipeline: extract → Demucs → faster-whisper → pyannote → IndicTrans2 → Indic Parler-TTS → time-fit → mix. Writes `.bn.wav`, `.bn.srt`, `.bn.mkv` (Bangla + original tracks), `timings.json` and `lines.json`
`spike_player.py`	PyQt6 + mpv player; plays the video with the Bangla track, toggles Bangla / original audio and Bangla subtitles
`requirements.txt` / `README.md`	Install steps per OS, gated-model sign-up, how to record each experiment

How to run

1. Install ffmpeg, libmpv and PyTorch, then `pip install -r requirements.txt`.
2. Accept the gated model pages on Hugging Face and run once with `--hf-token` to download models.
3. Turn the network off and run `python dub_pipeline.py clip.mp4 --offline`.
4. Play with `python spike_player.py clip.mp4`.

Tested so far: both scripts compile; mixing, time-stretch, Bangla subtitles and the two-track MKV export were run on a synthetic clip and work. The model stages have not run yet. Expect small library API fixes on the first real run.

Design choices in the code

- Non-English speech is translated to English by Whisper first, then English → Bangla by IndicTrans2.
- Each new speaker gets the next voice from Arjun, Aditi, Tapan, Rashmi, Arnav, Riya.
- A Bangla line longer than its slot is sped up, capped at 1.35×; lines at the cap are counted in `timings.json` for E7.

Go / no-go rules

Outcome	Decision
E1 passes and no other experiment hits its rework condition	**Go** to step 4, Development
E1 fails after one tuning retry	**Rework:** back to step 1 (see Rework Plan)
E2 fails, a local LLM passes	Go, with a local LLM for translation (raises hardware needs)
E5 under 80% after tuning	**Rework:** narrow BR-05 further, or ship the fallback release
E3, E4, E6, E7, E8 or E9 fail	Go with the BRD v1.1 fallback listed in the experiments table
2 rework loops on the same experiment	Product owner decides: fallback release (single-speaker dub + subtitles) or not feasible

Next step: run the prototype on the three test machines, fill the Results column above, then apply these rules.

Sources

- faster-whisper (GitHub)
- pyannote speaker-diarization-3.1 model card
- IndicTrans2 (GitHub)
- IndicTrans2 en-indic-dist-200M model card
- NLLB-200 (fairseq README)
- NLLB-200 distilled 600M model card
- Indic Parler-TTS model card
- bn-parler-tts model card
- SUBAK.KO dataset card
- RegSpeech12 paper (SUBAK.KO speaker regions)

Change log

Version	Date	Change
1.1	2026-10-29	Aligned with BRD v1.1 and the Rework Plan: 4-speaker test clip, Hindi as guaranteed language, new E9 for Bangladeshi vocabulary
1.1	2026-10-29	Each experiment now names its fallback; only E1 (and E5 under 80%) forces rework
1.0	2026-10-29	First version